

RIPHAH INTERNATIONAL UNIVERSITY
GULBERG GREEN CAMPUS



THE CLOSET FASHION ASSISTANT

AI-Based Smart Outfit Recommendation System

Submitted By

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1. EXECUTIVE SUMMARY

The Closet Fashion Assistant is an AI-powered intelligent outfit recommendation system designed to solve the daily challenge of deciding what to wear. The system acts as a Utility-Based Agent that considers multiple conflicting criteria including weather conditions, occasion suitability, personal style preferences, and outfit repetition avoidance.

Key Features Implemented:

Feature	Description
Real-Time Weather Integration	Fetches live weather data via OpenWeatherMap API
KNN Machine Learning	Classifies outfit acceptance based on learned patterns
Occasion-Based Filtering	Supports 6 occasion types with filtering
Smart Utility Scoring	Weighted scoring (Weather 35%, Occasion 30%, Style 20%, Diversity 15%)
A* Search Algorithm	Optimal search from 4.6M+ outfit combinations
Wear History Tracking	JSON-based storage of worn outfits
Modern GUI	Tkinter interface with medals and progress bars

2. PROBLEM DEFINITION

2.1 Real-World Problem

In today's fast-paced lifestyle, individuals spend significant time deciding what to wear daily. People face difficulty selecting outfits matching multiple criteria:

Challenge	Description
Weather Conditions	Seasonal and daily changes affect outfit suitability
Occasion Appropriateness	Different events require different dress codes
Personal Style	Individual preferences vary greatly
Outfit Repetition	Wearing same outfits frequently is undesirable

2.2 Formal Problem Formulation

Component	Definition
States	All possible outfit combinations (top + bottom + shoes)
Initial State	User wardrobe + current weather + selected occasion

Actions	Recommend outfit, accept/reject, mark as worn
Goal	Recommend most suitable outfit maximizing satisfaction
Cost	Penalty for weather/occasion/style mismatches and repetition

3. AGENT DESIGN & ARCHITECTURE

3.1 Utility-Based Agent

The system optimizes multiple criteria: weather match, occasion suitability, style preference, and outfit diversity. A utility-based approach enables simultaneous consideration of these conflicting objectives.

4. PEAS ANALYSIS

Component	Description
Performance Measure	High user acceptance, weather/occasion accuracy, minimal repetition, time saved
Environment	User's desktop application, digital wardrobe (CSV), real-time weather
Actuators	Display outfit recommendations, show alternatives, update wear history
Sensors	Weather API, user input, wardrobe dataset, wear history log

5. ENVIRONMENT PROPERTIES

Property	Category	Justification
Observability	Partially Observable	Agent only knows uploaded clothes from dataset
Dynamics	Stochastic	Weather can change; user may reject suggestions
Episodicity	Sequential	Each recommendation affects future state (history updates)
Dynamics Type	Dynamic	Real-time weather updates; wear history changes
Discreteness	Discrete	Finite number of clothes and combinations

Agency	Single-agent	Personal stylist for one user only
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6. MACHINE LEARNING (KNN)

6.1 K-Nearest Neighbors

Why KNN? Interpretable decisions, no retraining required, handles multi-type features, lightweight performance, proven in recommendation systems.

6.2 Configuration

Parameter	Value
k (neighbors)	5
Distance Metric	Euclidean
Weighting	Uniform
Algorithm	Ball Tree

6.3 Performance Metrics

Metric	Value
Accuracy	51%
Precision	56%
Recall	71%
F1-Score	0.626
Cross-Validation Best	71%

7. A* SEARCH ALGORITHM

7.1 Algorithm Formulation

The system uses A* search to find optimal outfit combinations from 4.6M+ possibilities.

Component	In This Project
State	Complete outfit (Top + Bottom + Shoes)
Start State	Available tops, bottoms, shoes from dataset
Goal State	Outfit with maximum utility score

Cost Function $g(n)$	Negative utility score (minimization)
Heuristic $h(n)$	0 (admissible - never overestimates)
Evaluation $f(n)$	$f(n) = g(n) + h(n) = \text{-utility}$

7.2 Search Space

Total combinations = Tops × Bottoms × Shoes = 162 × 159 × 179 = 4,610,682 possible outfits

8. UTILITY FUNCTION

Utility = (0.35 × Weather_Score) + (0.30 × Occasion_Score) + (0.20 × Style_Score) + (0.15 × Diversity_Score)

8.1 Score Components

Component	Weight	What it Measures
Weather Score	35%	How well outfit matches current temperature
Occasion Score	30%	Suitability for selected event
Style Score	20%	Similarity to user's taste (from KNN)
Diversity Score	15%	Penalty for recently worn items

9. DATASET DESCRIPTION

9.1 Dataset Overview

Property	Value
Name	closet_fashion_dataset.csv
Total Records	500 outfit items
Features	17 attributes
Target Variable	outfit_accepted (Binary: 0 or 1)
Positive Class	58% (289 accepted)
Negative Class	42% (211 rejected)

10. TECHNOLOGY STACK

Component	Technology	Purpose
Language	Python 3.14	Core development
ML Library	scikit-learn 1.8.0	KNN implementation
Data Processing	pandas 3.0.2, numpy 2.4.2	Data manipulation
GUI Framework	Tkinter	User interface
Weather API	OpenWeatherMap	Real-time weather data
Data Storage	JSON, CSV	Wear history & dataset

11. TESTING & RESULTS

All test cases passed successfully:

Test Case	Scenario	Status
TC-01	User enters valid city	✓ Pass
TC-02	User enters invalid city	✓ Pass
TC-03	Select Casual occasion	✓ Pass
TC-04	Select Formal occasion	✓ Pass
TC-05	Click Get Recommendations	✓ Pass
TC-06	Click Get More	✓ Pass
TC-07	Click Wear	✓ Pass
TC-08	Verify weather integration	✓ Pass
TC-09	Clear history	✓ Pass

12. ETHICAL & SOCIAL IMPACT

12.1 Risk Mitigation

Risk	Description	Mitigation
Privacy	Wardrobe data is personal	Local storage only, no cloud upload
Bias	Recommendations may favor certain styles	User controls occasion selection

Data Security	Wear history stored locally	JSON file on user device only
Over-reliance	Users may stop thinking independently	System recommends, not dictates

13. CONCLUSION

The Closet Fashion Assistant successfully implements a complete Utility-Based AI Agent that provides intelligent outfit recommendations by considering multiple criteria:

Key Achievements:

- ✧ Agent Design & PEAS Analysis - Complete
- ✧ KNN ML Implementation - Complete
- ✧ A* Search Algorithm - Complete
- ✧ GUI with all features - Complete
- ✧ Weather API Integration - Complete
- ✧ Wear History Tracking - Complete
- ✧ Working Demo - Complete